# Ivan Sepulveda

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## PROFESSIONAL SUMMARY

Analytics leader with 6+ years building predictive models, KPI frameworks, and automated SQL/Python reporting pipelines that drive operational and risk decisions. Experienced translating cross-functional business needs into scalable monitoring and forecasting solutions, partnering with engineering and business stakeholders, and presenting findings to senior leadership. Brings a rigorous quantitative foundation and full-stack data fluency to complex analytical problems.

## EDUCATION

**University of San Francisco**
*Bachelor of Science - Physics*

## EXPERIENCE

**Manager of Analytics, Counterparty Risk** **February 2026 – Present**
*Aqua Finance, Inc.* *Remote*

- Built counterparty risk frameworks, KPIs, early-warning indicators, and predictive models to monitor credit and fraud risk across dealers and applications
- Manage and mentor a Junior Analyst, and partner with Credit, Operations, Sales, Underwriting, and Data Engineering to translate business needs into scalable risk monitoring solutions
- Develop automated SQL/Python analytics pipelines and reporting, improving timeliness, consistency, and scalability of portfolio risk monitoring

**Sr. Risk Analyst** **June 2021 – February 2026**
*Solar Mosaic LLC* *Remote/Oakland, CA*

- Developed SQL/Gephi network analysis using IP, geographic, and business-registration data to identify risk and fraud among connected entities
- Built an XGBoost dealer offboarding model to predict each dealer's probability of becoming offboarded at different time horizons (3M, 6M, 12M)
- Implemented dynamic partner risk scorecards, daily-updating schemas, alerts, and data initiatives with Data Engineering using DBT, Apache Airflow, AWS, and Snowflake

**Lead Software Developer (Contract)** **October 2019 – June 2021**
*L-Egant Solutions, LLC* *Irvine, CA*

- Developed scripts and GUI software for high-frame-rate camera operation, video capture, image algebra, pixel-intensity analysis, and two-dimensional Gaussian characterization

**Data Analyst, Portfolio & Servicing Risk** **August 2019 – June 2021**
*Mechanics Bank Auto Finance* *Irvine, CA*

- Applied K-means, hierarchical clustering, and Random Forest models to segment collection agents, analyze credit metrics, inform credit policy, and forecast portfolio losses

## PROJECTS

**when-where** travel.iesepulveda.com
*Full-Stack Travel Recommendation Engine*

- **Data Sources:** REST APIs (World Bank Data360, Eurostat, Open-Meteo, Overpass) and web scraping (MICHELIN Guide) to build destination-scoring datasets
- **Pipeline:** Python ETL to clean, merge, and reconcile heterogeneous structured/unstructured data (weather, economic, crime, tourism) across inconsistent country/city coding schemes

**dft (Duty Free Tracker)** dft.iesepulveda.com
*Full-Stack Airport Shopping Data Platform*

- **Frontend:** Next.js/TypeScript — user login, favorites, interactive maps, image upload
- **Backend:** Supabase/Postgres — relational schema (Airport, Store, Sighting, Product, Brand)

## SKILLS

**Languages & Libraries:** Python, SQL, R, LaTeX, Java, Matplotlib, Mayavi, ggplot2, Tableau, pandas, scikit-learn
**Tools & Platforms:** DBT, Airflow, Docker, AWS, Snowflake, GitHub, Bitbucket, Gephi
**Methods:** Web Scraping, Network Analysis, Monte Carlo Simulations, XGBoost, Random Forest, KNN/K-Means

## PUBLICATIONS

- C. T. Chavez et al., "Measurement of 2D density profiles using a second-harmonic, dispersion interferometer," https://doi.org/10.1063/5.0119896
- M. Nikolic et al., "Applicability of optical emission spectroscopy techniques for characterization of Ar and Ar/O2 discharges," https://dx.doi.org/10.1088/1361-6463/abf61c